

Applied Mathematics I

Chapter one: Matrices and Determinant

Matrices

□ **Definition** : A **matrix** is a rectangular array of numbers, usually presented in the following form:

$$A = \begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \cdots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix}$$

□ The horizontal arrangement in the array is called row while the vertical arrangement is called column.

□ Note : The element a_{ij} , called the ij -entry appear in row i and column j . We denote such a matrix by simply writing $A = [a_{ij}]$.

Cont.

- A matrix with m rows and n columns is called an m by n matrix, written $m \times n$.
- The pair of numbers m and n is called the size of the matrix.
- **Definition:** Two matrices A and B are equal, written $A = B$, if and only if A and B have the same size and all their corresponding elements are equal.
- **Example:** Find x, y, z, t such that
$$\begin{bmatrix} x + y & 2z + t \\ x - y & z - t \end{bmatrix} = \begin{bmatrix} 3 & 7 \\ 1 & 5 \end{bmatrix}.$$

Algebra of Matrices

□ If A & B are both $m \times n$ matrices then $A \pm B$ is a new matrix that is obtained by adding /subtracting the corresponding entries from each other. $A \pm B = [a_{ij} \pm b_{ij}]$

□ **Example:** For the following matrices perform the indicated operations, if possible.

$$A = \begin{bmatrix} 2 & 0 & -3 & 2 \\ -1 & 8 & 10 & -5 \end{bmatrix} \quad B = \begin{bmatrix} 0 & -4 & -7 & 2 \\ 12 & 3 & 7 & 9 \end{bmatrix}$$

$$C = \begin{bmatrix} 2 & 0 & 2 \\ -4 & 9 & 5 \\ 6 & 0 & -6 \end{bmatrix}$$

a) $A + B$ b) $B - A$ c) $A + C$

Cont.

□ **Definition:** The product of the matrix A by a scalar c , written cA , is the matrix obtained by multiplying each element of A by c . That is,

$$cA = \begin{bmatrix} ca_{11} & ca_{12} & \cdots & ca_{1n} \\ ca_{21} & ca_{22} & \cdots & ca_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ ca_{m1} & ca_{m2} & \cdots & ca_{mn} \end{bmatrix}$$

□ **Example:** Given the matrices

$$\square A = \begin{bmatrix} 0 & 9 \\ 2 & -3 \\ -1 & 1 \end{bmatrix}, B = \begin{bmatrix} 8 & 1 \\ -7 & 0 \\ 4 & -1 \end{bmatrix}, C = \begin{bmatrix} 2 & 3 \\ -2 & 5 \\ 10 & -6 \end{bmatrix}$$

Compute $3A + 2B - \frac{1}{2}C$

Cont.

□ **Definition:** If A is an $m \times p$ matrix and B is a $p \times n$ matrix then the product is a new matrix with size $m \times n$ whose ij^{th} entry is found by multiplying row i of A times column j of B .

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1p} \\ a_{21} & a_{22} & \cdots & a_{2p} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mp} \end{bmatrix} \begin{bmatrix} b_{11} & b_{12} & \cdots & b_{1n} \\ b_{21} & b_{22} & \cdots & b_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ b_{p1} & b_{p2} & \cdots & b_{pn} \end{bmatrix}$$

$$= \begin{bmatrix} c_{11} & c_{12} & \cdots & c_{1n} \\ c_{21} & c_{22} & \cdots & c_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ c_{m1} & c_{m2} & \cdots & c_{mn} \end{bmatrix}$$

Cont.

where $c_{ij} = a_{i1}b_{1j} + a_{i2}b_{2j} + \cdots + a_{ip}b_{pj} = \sum_{k=1}^p a_{ik}b_{kj}$.

□ **Example:** compute AC and CA for the following two matrices, if possible.

$$A = \begin{bmatrix} 1 & -3 & 0 & 4 \\ -2 & 5 & -8 & 9 \end{bmatrix} \quad C = \begin{bmatrix} 8 & 5 & 3 \\ -3 & 10 & 2 \\ 2 & 0 & -4 \\ -1 & -7 & 5 \end{bmatrix}$$

□ The product AB is not defined if A is an $m \times p$ and B is a $q \times n$ matrix, where $p \neq q$.

□ **Definition:** A matrix obtained by interchanging rows and columns of A is called **transpose** of A and is denoted by A^t (or A^T).

□ **Example :** Let $A = \begin{bmatrix} 7 & -2 & 6 \\ 1 & 0 & 3 \end{bmatrix}$. Then $A^t = \begin{bmatrix} 7 & 1 \\ -2 & 0 \\ 6 & 3 \end{bmatrix}$

Properties of transpose

□ Let A and B be matrices and let k be a scalar. Whenever the sum and product are defined, then:

$$i. (A + B)^t = A^t + B^t$$

$$iii. (kA)^t = kA^t,$$

$$ii. (A^t)^t = A$$

$$iv. (AB)^t = B^t A^t$$

□ **Definition :** Let $A = [a_{ij}]$ be a square matrix. Then A is said to be

✓ *symmetric* if $A^t = A$. i.e. $a_{ij} = a_{ji}$.

✓ *skew – symmetric* if $A^t = -A$. i.e. $a_{ij} = -a_{ji}$.

□ **Example:**

$$\text{Let } A = \begin{bmatrix} 1 & -1 & 2 \\ -1 & 0 & 3 \\ 2 & 3 & 7 \end{bmatrix}, B = \begin{bmatrix} 0 & 3 & -4 \\ -3 & 0 & 5 \\ 4 & -5 & 0 \end{bmatrix}, C = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

Cont.

□Solution:

Since $A^t = A$. Thus A is symmetric.

$B^t = -B$. Thus B is skew- symmetric.

Since C is not square, C is neither symmetric nor skew-symmetric.

□**Remark** : Let A be a square matrix. Then

✓ $A + A^t$ is symmetric

✓ $A - A^t$ is skew-symmetric

✓ A can be written as a sum of a symmetric and skew-symmetric.

Cont.

□ **Example:** Express the given square matrix A as a sum of symmetric and skew symmetric, where

$$A = \begin{bmatrix} 1 & 3 & 2 \\ 4 & 0 & -3 \\ 1 & 2 & 5 \end{bmatrix}.$$

Solution: let $B = \frac{1}{2}(A + A^t)$ and $C = \frac{1}{2}(A - A^t)$

$$B = \frac{1}{2} \begin{bmatrix} 2 & 7 & 3 \\ 7 & 0 & -1 \\ 3 & -1 & 10 \end{bmatrix} \text{ is symmetric and } C = \frac{1}{2} \begin{bmatrix} 0 & -1 & 1 \\ 1 & 0 & -5 \\ -1 & 5 & 0 \end{bmatrix} \text{ skew-}$$

symmetric . So $B + C = A$

Types of matrices

- ❑ Row matrix: is a matrix that has only one row.
- ❑ Column matrix : is a matrix that has only one column.
- ❑ Zero matrix: All entries of matrix are zero.
- ❑ Square matrix : A matrix in which the number of rows & the number of Columns are equal .An $n \times n$ matrix is called square matrix of order n .
- ❑ The entries $a_{11}, a_{22}, \dots a_{nn}$ are called diagonal entries.
- ❑ The Diagonal containing each entry $a_{11}, a_{22}, \dots a_{nn}$ is called main diagonal or principal diagonal

Cont.

□ The sum of the diagonal entries of a square matrix A is called **trace** of A . Denoted by $tr(A)$.

$$tr(A) = a_{11} + a_{22} + \cdots + a_{nn} = \sum_{i=1}^n a_{ii}$$

□ **Diagonal matrix:** a square matrix A is called diagonal matrix if and only if all of its non-diagonal entries are zero.

□ **Scalar matrix :** A diagonal matrix in which all diagonal entries are equal is called scalar Matrix.

□ **Identity matrix:** A scalar matrix with diagonal entry of 1 is called identity matrix denoted by I_n

Cont.

□ **Upper triangular matrix.** A square matrix in which all the elements below the main diagonal are zero .

i.e. a matrix of type:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \dots & a_{1n} \\ 0 & a_{22} & a_{23} & \dots & a_{2n} \\ 0 & 0 & a_{33} & \dots & a_{3n} \\ \dots & \dots & \dots & \dots & \dots \\ 0 & 0 & 0 & 0 & a_{nn} \end{bmatrix}$$

Cont.

❑ **Lower triangular matrix.** A square matrix in which all the elements above the main diagonal are zero .

i.e. a matrix of type

$$\begin{bmatrix} a_{11} & 0 & 0 & \dots & 0 \\ a_{21} & a_{22} & 0 & \dots & 0 \\ a_{31} & a_{32} & a_{33} & \dots & 0 \\ \dots & \dots & \dots & \dots & \dots \\ a_{n1} & a_{n2} & a_{n3} & \dots & a_{nn} \end{bmatrix}$$

❑ A matrix which is either lower or upper triangular matrix is called **triangular matrix**.

Cont.

□ **Example:** Let $A = \begin{bmatrix} 1 & 2 & 3 \\ a+1 & 2 & d \\ b-2 & 0 & c \end{bmatrix}$. Find a, b, c and d so that A is triangular.

□ **Power of matrix :** the power of matrix A is the n^{th} power of A .

$$A^0 = I, A^1 = A, A^2 = A.A, A^3 = A^2.A, \dots, A^{n+1} = A^n A,$$

□ **Example:** find the n^{th} power of $A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$

Cont.

$$\square \text{Solution: } A^2 = A.A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix}$$

$$A^3 = A^2.A = \begin{bmatrix} 2 & 2 \\ 2 & 2 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix}$$

$$A^4 = A^3.A = \begin{bmatrix} 4 & 4 \\ 4 & 4 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 8 & 8 \\ 8 & 8 \end{bmatrix}$$

$$\text{Hence } A^n = 2^{n-1} \begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix} = \begin{bmatrix} 2^{n-1} & 2^{n-1} \\ 2^{n-1} & 2^{n-1} \end{bmatrix}, \text{for } n \geq 2$$

Inverse of matrices

□ **Definition:** The inverse of square matrix A , denoted by A^{-1} is a matrix so that the product of A and A^{-1} is identity matrix. i.e. $AA^{-1} = A^{-1}A = I$

□ A square matrix that has an inverse is called invertible or nonsingular matrix.

□ A matrix that does not have inverse is called singular matrix.

□ **Example :** Suppose $A = \begin{bmatrix} 1 & 0 & 2 \\ 2 & -1 & 3 \\ 4 & 1 & 8 \end{bmatrix}$ and $B = \begin{bmatrix} -11 & 2 & 2 \\ -4 & 0 & 1 \\ 6 & -1 & -1 \end{bmatrix}$,

then show that $B = A^{-1}$

Properties of inverse

$$\square (A^t)^{-1} = (A^{-1})^t .$$

Because $A^t (A^{-1})^t = (A^{-1} A)^t = I^t = I$.

$$\square (A^{-1})^t = (A^t)^{-1} = A^{-1}, \text{ hence } A^{-1} \text{ is also symmetric.}$$

$\square$ The inverse of a matrix product is the reverse product of the inverses of the factors:

$$(AB)^{-1} = B^{-1}A^{-1}.$$

$\square$ Hence for more than two factors,
$$(ABC \dots PQ)^{-1} = Q^{-1}P^{-1} \dots C^{-1}B^{-1}A^{-1}$$

Elementary row operations

□ **Definition** : Let A be an $m \times n$ matrix. The **elementary row operations** on A are:

✓ Interchange the i^{th} row and the j^{th} row:

$$R_i \leftrightarrow R_j$$

✓ Multiply the i^{th} row by a non zero scalar k :

$$R_i \rightarrow kR_i (k \neq 0)$$

✓ Replace the i^{th} row by k times the j^{th} row plus the i^{th} row:

$$R_i \rightarrow R_i + kR_j.$$

Row-echelon form(REF) and Reduced Row-echelon form(RREF)

□ **Definition:** A matrix is said to be row- echelon if and only if it satisfies:

- i. All zero rows (if any) below all non-zero rows.
- ii. If two successive rows of the matrix are non-zero, then the row with more zeros is below the row with less zero. (zeros are counted from left to right in each row).
- iii. All other elements of the column below the leading entry are zero.(Leading entry is the left most non-zero entry of a non-zero row).

□ A matrix is **Reduced Row-echelon** if it satisfies the two additional conditions:

- ✓ The leading entry is 1
- ✓ All other elements of the column in which leading 1 occurs are zero.

Cont.

□ **Example:** Explain whether the following matrices are row-echelon , reduced row-echelon form or neither.

$$A = \begin{bmatrix} 2 & 0 & 3 \\ 0 & 5 & 0 \\ 0 & 0 & 2 \end{bmatrix}, B = \begin{bmatrix} 1 & 5 & 3 \\ 0 & 1 & 2 \\ 0 & 0 & 0 \end{bmatrix}, C = \begin{bmatrix} 2 & 7 & 0 \\ 0 & 0 & 1 \\ 0 & 2 & 3 \end{bmatrix}$$

Solution:

- A and B are in row-echelon, only B is in reduced row-echelon.
- C is neither row –echelon nor reduced row echelon form.

Cont.

□ A matrix can be changed into Row Reduced –echelon form by using elementary row operations.

□ **Example:** Transform the following matrix into Row Reduced-echelon form .

$$A = \begin{bmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{bmatrix}$$

□ **Solution:**

□ Step 1: Multiply the 1st row by $\frac{1}{2}$.

Cont.

$$\begin{bmatrix} 2 & 0 & 4 & 2 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{bmatrix} \xrightarrow{R_1 \rightarrow \frac{1}{2}R_1} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{bmatrix}$$

Step 2: Reduce all other elements in the first column to zero.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 3 & 2 & 6 & 5 \\ 5 & 2 & 10 & 7 \\ 0 & 3 & 2 & 5 \end{bmatrix} \xrightarrow{\begin{matrix} R_2 \rightarrow R_2 - 3R_1 \\ R_3 \rightarrow R_3 - 5R_1 \end{matrix}} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 3 & 2 & 5 \end{bmatrix}$$

Cont.

□ Step 3: Now since the first nonzero element in the 2nd row is 2, then multiply the 2nd row by $\frac{1}{2}$.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 2 & 0 & 2 \\ 0 & 3 & 2 & 5 \end{bmatrix} \xrightarrow{\frac{1}{2}R_2 \rightarrow R_2} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 3 & 2 & 5 \end{bmatrix}.$$

□ Step 4: reduce all other elements in the second column to zero.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 2 & 0 & 2 \\ 0 & 3 & 2 & 5 \end{bmatrix} \xrightarrow{\begin{matrix} R_3 \rightarrow R_3 - 2R_2 \\ R_4 \rightarrow R_4 - 3R_2 \end{matrix}} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 \end{bmatrix}$$

Cont.

□ Step 5: Perform $R_3 \leftrightarrow R_4$ to bring the zero row to the last.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 2 & 2 \end{bmatrix} \xrightarrow{R_3 \leftrightarrow R_4} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

□ Step 6: Multiply the third row by $\frac{1}{2}$.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 2 & 2 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{\frac{1}{2}R_3 \rightarrow R_3} \begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Cont.

Step 7: Reduce all other elements in the third column to zero.

$$\begin{bmatrix} 1 & 0 & 2 & 1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix} \xrightarrow{R_1 \rightarrow R_1 - 2R_3} \begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

□ Therefore $\begin{bmatrix} 1 & 0 & 0 & -1 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{bmatrix}$ is a row reduced echelon matrix.

Rank of matrix

❑ **Definition:** Let A be an $m \times n$ arbitrary matrix. The number of non-zero rows in REF(RREF) of A is called rank of A . denoted by $\text{rank}(A)$

❑ **Example:** Determine the rank of the matrix

$$A = \begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 0 & 0 & 5 & 8 \\ 3 & 6 & 6 & 3 \end{bmatrix}$$

$$B = \begin{bmatrix} 3 & 1 & 2 & 5 \\ 1 & 2 & -1 & 2 \\ 4 & 3 & 1 & 7 \end{bmatrix}$$

❑ **Solution:**

$$\begin{bmatrix} 1 & 2 & 1 & 0 \\ 2 & 4 & 8 & 6 \\ 0 & 0 & 5 & 8 \\ 3 & 6 & 6 & 3 \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_4 \rightarrow R_4 - 3R_1}} \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 5 & 8 \\ 0 & 0 & 3 & 3 \end{bmatrix} \xrightarrow{\substack{R_3 \rightarrow R_3 - \frac{5}{6}R_2 \\ R_4 \rightarrow R_4 - \frac{1}{2}R_2}} \begin{bmatrix} 1 & 2 & 1 & 0 \\ 0 & 0 & 6 & 6 \\ 0 & 0 & 0 & 3 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

The number of nonzero rows in REF of A is 3. Therefore $\text{rank}(A)=3$

Solving system of linear Equations by Gaussian eliminations

□ **Definition:** A system of m linear equations in n unknowns $x_1, x_2, \dots, x_n$ is of the form:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots \\a_{m1}x_1 + a_{m2}x_2 + \cdots + a_{mn}x_n &= b_m.\end{aligned}$$

□ if all b_i 's are zero the system is called homogeneous system.

□ if at least one of b_i 's is non-zero the system is called non-homogeneous system

Cont.

□ The above system is equivalent to the following matrix equation:

$$\begin{bmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \cdots & \cdots & \cdots & \cdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{bmatrix} \text{ or simply } AX = B.$$

where A is coefficient matrix, X is variable matrix and B is constant matrix.

□ $(A|B)$ is called augmented matrix.

Solution of a system of linear equations and number of Solutions

□ **Definition:** The value of variables $x_1, x_2, \dots, x_n$ that satisfy all the equations of the system are called solution of the system.

□ The system is consistent if it has at least one solution.

□ If the system has no solution it is called inconsistent.

□ **Theorem (the rank test for number of Solution)**

Suppose $AX = B$ is a linear equation where A is $m \times n$ matrix. Then

✓ If $\text{rank}(A|B) = \text{rank}(A) = n$, then the system has unique solution.

✓ If $\text{rank}(A|B) = \text{rank}(A) = r < n$, then the system has more than one solution.

✓ If $\text{rank}(A|B) \neq \text{rank}(A)$ then the system has no solution.

Cont.

□Examples:

1. Determine the number of solutions of the following:

$$x_1 + 2x_2 - x_3 = 10$$

$$-x_1 + x_2 + 2x_3 = 2$$

$$2x_1 + x_2 - 3x_3 = 2$$

2. Determine the values of a and b so that the following system will have unique solution, many solution and no solution.

$$\begin{cases} x - 2y + 3z = 4 \\ 2x - 3y + az = 5 \\ 3x - 4y + 5z = b \end{cases}$$

Cont.

□ **Solution:**

□ The coefficient matrix $A = \begin{bmatrix} 1 & 2 & -1 \\ -1 & 1 & 2 \\ 2 & 1 & -3 \end{bmatrix}$ and augmented matrix

$$(A|B) = \begin{bmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 2 \end{bmatrix}$$

□ Apply elementary row operation to augmented matrix.

Cont.

$$\text{That is } \begin{bmatrix} 1 & 2 & -1 & 10 \\ -1 & 1 & 2 & 2 \\ 2 & 1 & -3 & 2 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & -3 & -1 & -18 \end{bmatrix}$$

$$\begin{bmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & -3 & -1 & -18 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 + R_2} \begin{bmatrix} 1 & 2 & -1 & 10 \\ 0 & 3 & 1 & 12 \\ 0 & 0 & 0 & 6 \end{bmatrix}$$

$$\text{rank}(A) = 2, \text{rank}(A|B) = 3$$

□ This implies that $\text{rank}(A|B) \neq \text{rank}(A)$.

□ Thus by rank test the system has no solution.

Cont.

$$\square 2. \text{Coefficient matrix} = \begin{bmatrix} 1 & -2 & 3 \\ 2 & -3 & a \\ 3 & -4 & 5 \end{bmatrix}, \text{Constant Matrix} = \begin{bmatrix} 4 \\ 5 \\ b \end{bmatrix}$$

Augmented matrix = $\begin{bmatrix} 1 & -2 & 3 & 4 \\ 2 & -3 & a & 5 \\ 3 & -4 & 5 & b \end{bmatrix}$. Change the augmented matrix into row echelon form.

$$\begin{bmatrix} 1 & -2 & 3 & 4 \\ 2 & -3 & a & 5 \\ 3 & -4 & 5 & b \end{bmatrix} \xrightarrow{\substack{R_2 \rightarrow R_2 - 2R_1 \\ R_3 \rightarrow R_3 - 2R_1}} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & a-6 & -3 \\ 0 & 2 & -4 & b-12 \end{bmatrix}$$

Cont.

$$\begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & a-6 & -3 \\ 0 & 2 & -4 & b-12 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 2R_2} \begin{bmatrix} 1 & -2 & 3 & 4 \\ 0 & 1 & a-6 & -3 \\ 0 & 0 & -2a+8 & b-6 \end{bmatrix}$$

it is in row echelon form.

□ Has unique solution if $\text{rank}(A|B) = \text{rank}(A) = 4$

$\Rightarrow -2a + 8 \neq 0$ and $b \in \mathbb{R}$

$\Rightarrow a \neq 4$ and $b \in \mathbb{R}$

□ Has many solution if $\text{rank}(A|B) = \text{rank}(A) < 4$

$\Rightarrow -2a + 8 = 0$ and $b - 6 = 0$

$\Rightarrow a = 4$ and $b = 6$

□ Has no solution if $\text{rank}(A|B) \neq \text{rank}(A)$

$\Rightarrow -2a + 8 = 0$ and $b - 6 \neq 0$

$\Rightarrow a = 4$ and $b \neq 6$

Gaussian elimination method

- ❑ To solve a system of linear equation by Gaussian elimination method:
- Write a system of linear equations as an augmented matrix.
- Perform the elementary row operations to put the matrix into row-echelon form.
- Convert the matrix back into a system of linear equations.
- Use back substitution to obtain all the answers.

Cont.

Examples: solve the following system of linear equation by using Gaussian elimination method .

$$x_1 - 2x_2 + 3x_3 = -2$$

1. $-x_1 + x_2 - 2x_3 = 3$

$$2x_1 - x_2 + x_3 = 1$$

2.

$$x_1 + x_2 - x_3 + 3x_4 = 4$$

$$x_1 + 2x_2 + 3x_3 - x_4 = 3$$

$$5x_1 + 7x_2 + 4x_3 + x_4 = 5$$

$$x_1 - 2x_2 + x_3 = -2$$

3. $-x_1 + x_2 - 2x_3 = 3$

$$2x_1 - 3x_2 + 3x_3 = -7$$

Cont.

□ Solution: 1. coefficient matrix = $\begin{bmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 2 & -1 & 1 \end{bmatrix}$,

□ constant matrix = $\begin{bmatrix} -2 \\ 3 \\ 1 \end{bmatrix}$ and augmented

matrix = $\begin{bmatrix} 1 & -2 & 3 & -2 \\ -1 & 1 & -2 & 3 \\ 2 & -1 & 1 & 1 \end{bmatrix}$

□ Change augmented matrix to row echelon form.

Cont.

$$\begin{bmatrix} 1 & -2 & 3 & -2 \\ -1 & 1 & -2 & 3 \\ 2 & -1 & 1 & 1 \end{bmatrix} \xrightarrow[R_3 \rightarrow R_3 - 2R_1]{R_2 \rightarrow R_2 + R_1} \begin{bmatrix} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 3 & -5 & 5 \end{bmatrix} \xrightarrow{R_3 \rightarrow R_3 - 3R_2} \begin{bmatrix} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & -2 & 8 \end{bmatrix}$$

□ Back substitution

$$\begin{bmatrix} 1 & -2 & 3 \\ 0 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} -2 \\ 1 \\ 8 \end{bmatrix}$$

□ Solve for x_1, x_2 and x_3

□ $x_1 - 2x_2 + 3x_3 = -2$

• $-x_2 + 3x_3 = 1$

• $-2x_3 = 8 \Rightarrow x_3 = -4, x_2 = -13, x_1 = -16$

Determinants

Definitions and properties of determinant

□ Determinant of square matrix A is denoted as $\det A$ or $|A|$.

□ The determinants of order one (1×1) matrix $A = (a_{11})$ is the scalar a_{11} it self, that is, $\det(A) = |a_{11}| = a_{11}$.

□ Determinant of order two matrix $A = \begin{bmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{bmatrix}$ is defined as

$$\begin{vmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{vmatrix} = a_{11}a_{22} - a_{21}a_{12}$$

□ **Definition :** Consider an arbitrary 3×3 matrix $A = (a_{ij})$. The determinant of A is defined as follows:

$$\text{Det}(A) = \begin{vmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{vmatrix}$$

Cont.

$$= a_{11} \begin{vmatrix} a_{22} & a_{23} \\ a_{32} & a_{33} \end{vmatrix} - a_{12} \begin{vmatrix} a_{21} & a_{23} \\ a_{31} & a_{33} \end{vmatrix} + a_{13} \begin{vmatrix} a_{21} & a_{22} \\ a_{31} & a_{32} \end{vmatrix}$$

□ **Note:** Let A_{ij} be the matrix obtained from A by crossing out the i^{th} row and the j^{th} column of A . Then:

$$|A| = a_{11}|A_{11}| - a_{12}|A_{12}| + a_{13}|A_{13}|$$

Determinant of n Order

□ $\det(A) = (-1)^{i+1}a_{i1}|A_{i1}| + (-1)^{i+2}a_{i2}|A_{i2}| + (-1)^{i+n}a_{in}|A_{in}|$ that is, expansion according to the j^{th} column

□ **Example :** Let $B = \begin{bmatrix} 0 & 2 & 1 & 3 \\ 4 & 2 & -1 & 0 \\ 6 & 2 & 0 & 1 \\ 0 & 1 & 3 & 2 \end{bmatrix}$

Cont.

Solution:

□ Expand according to the 1st column, and then we have

$$\begin{aligned} \det B &= (-1)^{1+1}(0) \begin{vmatrix} 2 & -1 & 0 \\ 2 & 0 & 1 \\ 1 & 3 & 2 \end{vmatrix} + (-1)^{1+2}(4) \begin{vmatrix} 2 & 1 & 3 \\ 2 & 0 & 1 \\ 1 & 3 & 2 \end{vmatrix} \\ &+ (-1)^{1+3}(6) \begin{vmatrix} 2 & 1 & 3 \\ 2 & -1 & 0 \\ 1 & 3 & 2 \end{vmatrix} + (-1)^{1+4}(0) \begin{vmatrix} 2 & 1 & 3 \\ 2 & -1 & 0 \\ 2 & 0 & 1 \end{vmatrix}. \\ \det B &= 42. \end{aligned}$$

Properties of determinants

- Let A be a square matrix. Then $|A| = |A^t|$.
- If two rows or two columns, are interchanged the sign of the determinant is changed.
- **Example:** $\begin{vmatrix} 3 & 4 \\ 1 & -2 \end{vmatrix} = -\begin{vmatrix} 1 & -2 \\ 3 & 4 \end{vmatrix}$.
- If a row or column is changed by adding to or subtracting from its elements the corresponding elements of any other row or column the determinant remains unaltered.
- **Example:** $\begin{vmatrix} 3 & 4 \\ 1 & -2 \end{vmatrix} = \begin{vmatrix} 3+1 & 4-2 \\ 1 & -2 \end{vmatrix} = \begin{vmatrix} 4 & 2 \\ 1 & -2 \end{vmatrix} = -10$.
- If A has a row (column) of zeros, then $|A| = 0$.

Cont.

□ Example:
$$\begin{vmatrix} 2 & 3 & 4 \\ 0 & 0 & 0 \\ 1 & 5 & 6 \end{vmatrix} = 2 \begin{vmatrix} 0 & 0 \\ 5 & 6 \end{vmatrix} - 0 \begin{vmatrix} 3 & 4 \\ 5 & 6 \end{vmatrix} + \begin{vmatrix} 3 & 4 \\ 0 & 0 \end{vmatrix}$$
$$= 2(0) - 0(-2) + 0 = 0.$$

□ When at least one row (or column) of a matrix is a linear combination of the other rows (or columns) the determinant is zero.

Example :
$$\begin{vmatrix} 3 & 2 & 1 \\ 1 & 2 & -1 \\ 2 & -1 & 3 \end{vmatrix} = 0, \text{ column } 1 = \text{column } 2 + \text{column } 3.$$

□ If A is triangular, then $|A| =$ product of diagonal elements. Thus in particular, $|I| = 1$, where I is the identity matrix.

Cont.

□ If a row (or column) of A were multiplied by a scalar k , then $|B| = k|A|$.

Example: $\begin{vmatrix} 6 & 8 \\ 1 & -2 \end{vmatrix} = \begin{vmatrix} 2(3) & 2(4) \\ 1 & -2 \end{vmatrix} = 2 \begin{vmatrix} 3 & 4 \\ 1 & -2 \end{vmatrix} = 2(-10) = -20.$

□ (*Determinant of product of matrices*): $|AB| = |A||B|$.

□ If A is multiplied by the scalar k , then $|kA| = k^n |A|$

Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$ then $|4A| = 4^3 |A|$

Minors and cofactors

□ **Definition** : Consider an n -square matrix $A = [a_{ij}]$. Let A_{ij} denote the $(n - 1)$ square sub matrix of A obtained by deleting its i^{th} row and j^{th} column. Then

✓ $M_{ij} = |A_{ij}|$ is called the **minor** of the element a_{ij} of A

✓ $c_{ij} = (-1)^{i+j} |M_{ij}|$ is a **cofactor** of a_{ij}

✓ The **adjoint** of A , denoted by $adjA$, is the transpose of the matrix of cofactors of A . that is

$$✓ adjA = [c_{ij}]^t = \begin{bmatrix} c_{11} & c_{21} & \cdots & c_{n1} \\ c_{12} & c_{22} & \cdots & c_{n2} \\ \cdots & \cdots & \cdots & \cdots \\ c_{1n} & c_{2n} & \cdots & c_{nn} \end{bmatrix}$$

Cont.

□ If $\det A \neq 0$, then A is non-singular and $A^{-1} = \frac{1}{\det A} \text{adj} A$.

□ **Example:** Let $A = \begin{bmatrix} 2 & 3 & -4 \\ 0 & -4 & 2 \\ 1 & -1 & 5 \end{bmatrix}$, then find A^{-1}

□ **Solution:**

$$\begin{aligned} \bullet A_{11} &= + \begin{vmatrix} -4 & 2 \\ -1 & 5 \end{vmatrix} = -18, & A_{12} &= - \begin{vmatrix} 0 & 2 \\ 1 & 5 \end{vmatrix} = 2, & A_{13} &= + \begin{vmatrix} 0 & -4 \\ 1 & -1 \end{vmatrix} = 4 \\ \bullet A_{21} &= - \begin{vmatrix} 3 & -4 \\ -1 & 5 \end{vmatrix} = -11, & A_{22} &= + \begin{vmatrix} 2 & -4 \\ 1 & 5 \end{vmatrix} = 14, & A_{23} &= - \begin{vmatrix} 2 & 3 \\ 1 & -1 \end{vmatrix} = 5 \\ \bullet A_{31} &= + \begin{vmatrix} 3 & -4 \\ -4 & 2 \end{vmatrix} = -10, & A_{32} &= - \begin{vmatrix} 2 & -4 \\ 0 & 2 \end{vmatrix} = -4, & A_{33} &= + \begin{vmatrix} 2 & 3 \\ 0 & -4 \end{vmatrix} = -8 \end{aligned}$$

Cont.

□ The transpose of the above matrix of cofactors yields the classical adjoint of A, that is,

$$\text{adj}A = \begin{bmatrix} -18 & -11 & -10 \\ 2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix}$$

$$\det A = -22$$

$$\square A^{-1} = \frac{1}{\det A} \text{adj}A = -\frac{1}{22} \begin{bmatrix} -18 & -11 & -10 \\ 2 & 14 & -4 \\ 4 & 5 & -8 \end{bmatrix}$$

Cramer's Rule

□ Consider a system of n linear equations in n unknowns:

$$\begin{aligned}a_{11}x_1 + a_{12}x_2 + \cdots + a_{1n}x_n &= b_1 \\a_{21}x_1 + a_{22}x_2 + \cdots + a_{2n}x_n &= b_2 \\&\vdots\end{aligned}$$

$$a_{n1}x_1 + a_{n2}x_2 + \cdots + a_{nn}x_n = b_n.$$

□ Matrix form $AX = B$, where $A = (a_{ij})$ is a coefficient matrix and $B = (b_i)$ constant matrix.

□ Let A_i be the matrix obtained from A by replacing the i^{th} column of A by the column vector B .

□ Let $\Delta = \det(A)$ and let $\Delta_i = \det(A_i)$ for $i = 1, 2, \dots, n$.

Cont.

□ **Theorem :**(Cramer's rule): The above system has a unique solution if and only if $\Delta \neq 0$. In this case the unique solution is given by:

$$x_1 = \frac{\Delta_1}{\Delta}, x_2 = \frac{\Delta_2}{\Delta}, \dots, x_n = \frac{\Delta_n}{\Delta}.$$

□ **Example:**

Solve the following

$$\begin{cases} 2x + y - z = 3 \\ x + y + z = 1 \\ x - 2y - 3z = 4 \end{cases}$$

□ First compute the determinant Δ of the matrix of coefficients:

$$\Delta = \begin{vmatrix} 2 & 1 & -1 \\ 1 & 1 & 1 \\ 1 & -2 & -3 \end{vmatrix} = 5$$

Cont.

□ Since $\Delta \neq 0$, the system has a unique solution. To compute Δ_1, Δ_2 , and Δ_3 , replace the coefficients of x, y, z in the matrix of coefficients by the constant terms:

$$\Delta_1 = \begin{vmatrix} 3 & 1 & -1 \\ 1 & 1 & 1 \\ 4 & -2 & -3 \end{vmatrix} = -9 + 4 + 2 + 4 + 6 + 3 = 10$$

$$\Delta_2 = \begin{vmatrix} 2 & 3 & -1 \\ 1 & 1 & 1 \\ 1 & 4 & -3 \end{vmatrix} = -6 + 3 - 4 + 1 - 8 + 9 = -5$$

$$\Delta_3 = \begin{vmatrix} 2 & 1 & 3 \\ 1 & 1 & 1 \\ 1 & -2 & 4 \end{vmatrix} = 8 + 1 - 6 - 3 + 4 - 4 = 0$$

□ Thus the unique solution is $x = \frac{\Delta_1}{\Delta} = 2, y = \frac{\Delta_2}{\Delta} = -1, z = \frac{\Delta_3}{\Delta} = 0$.

Cramer's rule as test of solutions

- If $\Delta \neq 0$, then the system has unique solution
- If $\Delta = 0$ and at least one of $\Delta_i \neq 0$, then the system has no solution.
- If $\Delta = 0$ and $\Delta_i = 0$ for all $i = 1, 2, 3, \dots, n$ then the system has infinitely many solution (but it cannot be solved by Cramer's rule).
- **Example:** Using Cramer's rule, find the values of a and b for which the following system has unique, many and no solution.

$$\square \begin{cases} x - 2y + 3z = 4 \\ 2x - 3y + az = 5 \\ 3x - 4y + 5z = b \end{cases}$$

Cont.

□ **Solution:**

□ First compute the determinant Δ of the matrix of coefficients:

$$\Delta = \begin{vmatrix} 1 & -2 & 3 \\ 2 & -3 & a \\ 3 & -4 & 5 \end{vmatrix} = \begin{vmatrix} -3 & a \\ -4 & 5 \end{vmatrix} - 2 \begin{vmatrix} -2 & 3 \\ -4 & 5 \end{vmatrix} + 3 \begin{vmatrix} -2 & 3 \\ -3 & a \end{vmatrix} = 8 - 2a$$

$$\begin{aligned} \bullet \Delta_1 &= \begin{vmatrix} 4 & -2 & 3 \\ 5 & -3 & a \\ b & -4 & 5 \end{vmatrix} = 4 \begin{vmatrix} -3 & a \\ -4 & 5 \end{vmatrix} - 5 \begin{vmatrix} -2 & 3 \\ -4 & 5 \end{vmatrix} + b \begin{vmatrix} -2 & 3 \\ -3 & a \end{vmatrix} \\ &= 16a + 9b - 2ab - 70 \end{aligned}$$

Cont.

□ The system has unique solution if $\Delta \neq 0$

$$\Rightarrow 8 - 2a \neq 0$$

$$\Rightarrow 2a \neq 8$$

$$\Rightarrow a \neq 4 \text{ and } b \in \mathbb{R}$$

□ The system has many solution if $\Delta = 0$ and $\Delta_i = 0$

for all $i = 1, 2, 3, \dots, n$

$$\Rightarrow 8 - 2a = 0 \text{ and } 16a + 9b - 2ab - 70 = 0$$

$$\Rightarrow a = 4 \text{ and } 64 + 9b - 8b - 70 = 0$$

$$\Rightarrow a = 4 \text{ and } b = 6$$

Cont.

□ The system has no solution if $\Delta = 0$ and at least one of $\Delta_i \neq 0$

$$\Rightarrow 8 - 2a = 0 \text{ and } 16a + 9b - 2ab - 70 \neq 0$$

$$\Rightarrow a = 4 \text{ and } 64 + 9b - 8b - 70 \neq 0$$

$$\Rightarrow a = 4 \text{ and } b \neq 6$$

Review Exercise

1. Use Cramer's Rule to solve the system

$$2w - 6u - 6v = 4$$

$$3w - 2u + v = 3$$

$$w - u + v = 5$$

2. If $\begin{vmatrix} a & x & m \\ b & y & n \\ c & z & s \end{vmatrix} = 6$, then Find $\begin{vmatrix} a-m & -m & 4x \\ 2b-2n & -2n & 8y \\ 3c-3s & -3s & 12z \end{vmatrix}$

Continue

3. Consider the following system

$$\begin{aligned}x - 2y + bz &= 3 \\ ax + 2z &= 2 \\ 5x + 2y &= 1\end{aligned}$$

Find all rational numbers a and b for which the following system has

- i. No solution
- ii. Exactly one solution
- iii. Infinitely many solution

4. Solve the systems of equations

$$\begin{aligned}2x + 4y + 2z &= 4 \\ 6x + 12y + 2z &= 8 \\ x + 2y &= 1\end{aligned}$$

Continue

5. Let $\begin{cases} x - 2y + bz = 3 \\ ax + 2z = 2 \\ 5x + 2y = 2 \end{cases}$ be given. Find a and b for which the system has

exactly one solution.

6. Let $A = \begin{pmatrix} 1 & 2 \\ k & 4 \end{pmatrix}$

a. For what values of k is A symmetric?

b. For what values of k is A invertible?

c. For what values of k is $\text{rank}(A) = 1$?

Continue

7. By using property of determinant prove that

$$\begin{vmatrix} 1 & m & m^2 & m^3 \\ m^3 & 1 & m & m^2 \\ m^2 & m^3 & 1 & m \\ m & m^2 & m^3 & 1 \end{vmatrix} = (1 - m^4)^3$$

8. Let $A = \begin{bmatrix} 1 & 2 & 4 \\ 2 & 3 & 5 \\ 4 & k & 6 \end{bmatrix}$

- a) For what value of k the matrix A is invertible.
- b) For what value of k the matrix A is symmetric.